REG NO: 22100934340091

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION

MODULE NAME: SIGNAL AND SYSTEMS

MODULE CODE: ET 8211

TEST 1 DATE: 30th April, 2024

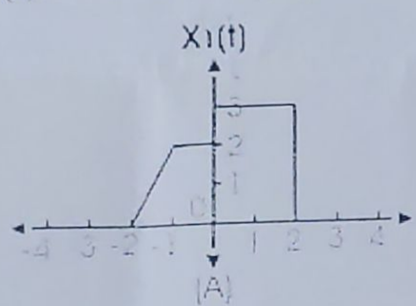
TIME: 1:30 Hours ANSWER ALL QUESTIONS

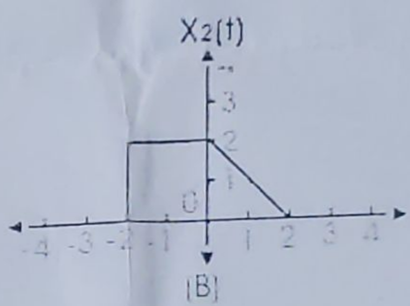
1. (a)Define the following, each give at least two examples (3 marks)
   (i) Signal (ii) System
   (b) What are the properties linear system should satisfy? (2 marks)
   (c) Differentiate between Causal and non-Causal systems (2 marks)

2. (a) Mention any two differences between Fourier series and Fourier transform (2 marks)
   (b)What are the types of Fourier series expansion? (3 marks)
   (c) Mention the conditions for Fourier series (3 marks)
   (d) Differentiate between Z and Laplace transform (2 marks)

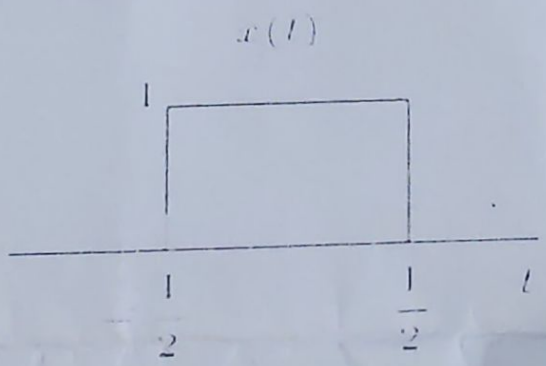
3. (a)Determine the system function of the discrete time system described by the difference equation (3 marks)
   $Y(n) - \frac{1}{2}y(n-1) + \neq \frac{1}{4}y(n-2) = x(n) - x(n-1)$
   (b) What are the Poles and Zeros of a system having the following transfer function? (3 marks)
   $H(z)=(1-z^{-2})/(1+1.3z^{-1}+0.36z^{-2})$

4. (a)Add the following signals (3 marks)





(b) Given the following signal



Perform the following and plot y(t)
   (i) $y(t) = x(3t + 2)$ (2 marks)
   (ii) $y(t) = x(2 - t/2)$ (2 marks)

$y(n) - \frac{1}{2}y(n-1) + \frac{1}{4}y(n-2) = x(n) - x(n-1).$